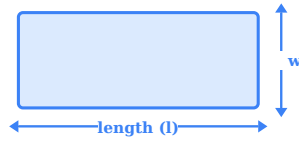


Area Cheat Sheet

6th Grade Math · Formulas, Diagrams & Real-World Examples

Rectangle

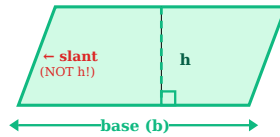


$$A = l \times w$$

✓ Multiply the two side lengths. That's it!

■ **At home:** Front door · TV screen · Phone · Notebook · Window · Table top · Picture frame

Parallelogram

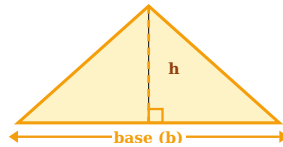


$$A = b \times h$$

△ h = dotted vertical line — NOT the slanted side!

■ **At home:** Side of a ramp or slide · Diamond-patterned wall tiles · Staircase railing · Like a rectangle pushed sideways!

Triangle

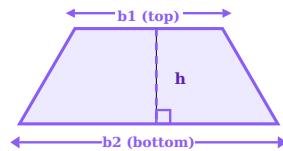


$$A = \frac{1}{2} \times b \times h$$

✓ Triangle = half a rectangle. ALWAYS divide by 2!

■ **At home:** Pizza slice · Sandwich cut corner-to-corner · Tortilla chip · Home roofline (side view) · Yield sign

Trapezoid



$$A = \frac{1}{2} \times (b1 + b2) \times h$$

✓ Add BOTH bases first, then $\times h$, then $\div 2$

■ **At home:** Bucket / pail (side view) · Lampshade · Bathtub (end view) · Paper cup opened flat

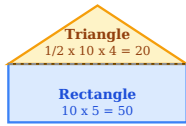
Top Mistakes to Avoid on the Test

- ✗ **Parallelogram & Triangle:** The slanted edge is NOT the height. Use the dotted perpendicular line with the little square corner $\square$ — that's your h.
- ✗ **Triangle:** Did you divide by 2? Every triangle = half a rectangle. The $\frac{1}{2}$ must always be in the formula.
- ✗ **Trapezoid:** Add $b1 + b2$ *first* before multiplying by height. Don't skip that step!
- ✗ **Units:** Write square units: cm^2 , ft^2 , in^2 , m^2 . Missing units = lost points!
- ✓ **The little square $\square$ on a diagram** marks a 90° right angle — that marks where h is perpendicular to the base.

Complex / Composite Shapes

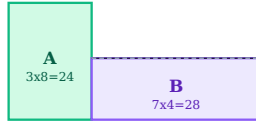
← **THE RULE:** Break the big shape into simple shapes → Find each area → ADD or SUBTRACT

+ ADD — "HOUSE SHAPE"



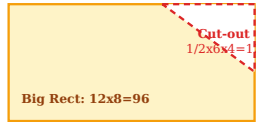
1. Rectangle: $10 \times 5 = 50 \text{ cm}^2$
2. Triangle: $\frac{1}{2} \times 10 \times 4 = 20 \text{ cm}^2$
3. Total: $50 + 20 = 70 \text{ cm}^2$

+ ADD — "L-SHAPE"



1. Area A: $3 \times 8 = 24 \text{ cm}^2$
2. Area B: $7 \times 4 = 28 \text{ cm}^2$
3. Total: $24 + 28 = 52 \text{ cm}^2$

— SUBTRACT — "CUTOUT"



1. Big rect: $12 \times 8 = 96 \text{ cm}^2$
2. Cut-out: $\frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$
3. Shaded: $96 - 12 = 84 \text{ cm}^2$

Practice Problems

Use the cheat sheet on page 1 to help. Show your work!

LOW — Warm Up

1

RECTANGLE

A classroom whiteboard is **12 ft long** and **4 ft tall**. What is the area of the whiteboard?

Area = _____

2

TRIANGLE

A triangle has a **base of 10 cm** and a **height of 6 cm**. What is its area?

Area = _____

MEDIUM — Level Up

3

PARALLELOGRAM

A parallelogram has a **base of 9 in**, a **perpendicular height of 5 in**, and a slant side of 7 in. What is the area? (*Careful — which measurement do you need?*)

Area = _____

4

TRAPEZOID

A trapezoid has a **top base (b1) of 6 ft**, a **bottom base (b2) of 10 ft**, and a **height of 4 ft**. What is its area?

Area = _____

5

RECTANGLE — DECIMALS

A rectangular swimming pool is **24 meters long** and **9.5 meters wide**. What is the area of the pool's surface?

Area = _____

6

TRIANGLE

A triangular road sign has a **base of 14 inches** and a **height of 12 inches**. What is the area of the sign?

Area = _____

HIGH — Challenge Round

7

COMPOSITE: RECTANGLE + TRIANGLE

A "house" shape has a rectangular base **8 m wide × 5 m tall**, and a triangular roof on top with **base 8 m and height 3 m**. What is the TOTAL area?

Total Area = _____

8

COMPOSITE: L-SHAPE

An L-shaped garden has **Rectangle A: 6 ft × 10 ft** and **Rectangle B: 4 ft × 5 ft**. What is the total area of the garden?

Total Area = _____

9

TRAPEZOID — WORD PROBLEM

A farmer's trapezoidal field has a **top edge of 35 m**, a **bottom edge of 55 m**, and a **perpendicular distance of 20 m** between them. How many square meters can the farmer plant?

Area = _____

10

WORK BACKWARDS — FIND THE MISSING MEASUREMENT

A parallelogram has an **area of 72 cm²** and a **base of 12 cm**. What is its **height**? (*Hint: $A = b \times h$, so solve for h .*)

Height = _____

Reveal only after your student has tried every problem!

1 · Rect
48 ft²

2 · Tri
30 cm²

3 · Para
45 in²

4 · Trap
32 ft²

5 · Rect
228 m²

6 · Tri
84 in²

7 · Comp
52 m²

8 · L-Shape
80 ft²

9 · Trap
900 m²

10 · h=?
6 cm